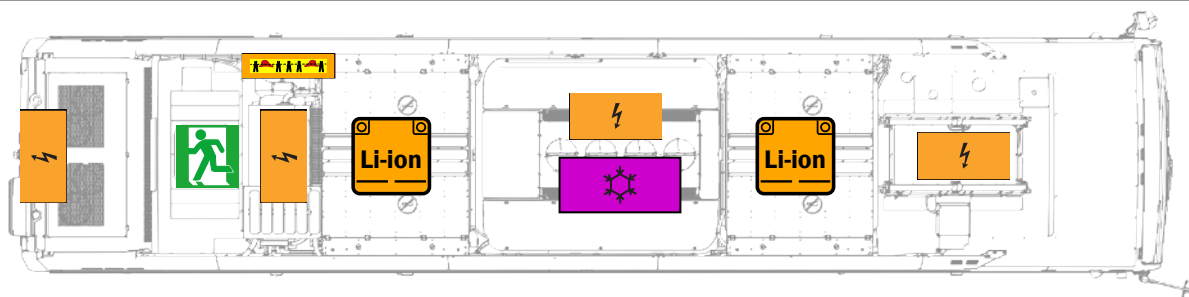


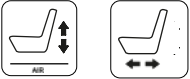
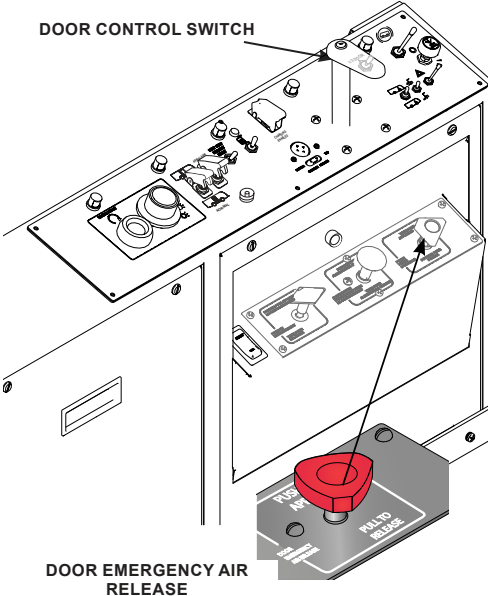
NOVABUS

NOVA BUS LFSE+ PRODUCTION START 2022



Emergency door opener	Emergency exits	High-voltage Lithium-ion battery	Low-voltage device that disconnects high voltage	High-voltage power cable	High-voltage components
Location to Cut High-Voltage Interlock Loop	Low-Voltage Battery	Device to Shut Down Power in Vehicle	Lifting Point	Seat Adjustment	Steering Wheel Tilt Control
Height Control	Air-Conditioning Unit	Air-Conditioning Line	Gas Strut Preloaded Spring	Air Tank	Diesel Fuel

	Identification number	Version number	Page number
	800077265	2024 August	1

1. Identification/Recognition			
 The Nova Bus LFSe+ is ONLY available with an electric drive. • No exhaust system • Low-voltage charging socket, rear streetside and curbside • High-voltage charging rails on the roof above driver • High-voltage battery state of charge (SOC) and power flow display devices on the dashboard • Orange high-voltage cables in the motor compartment and inside roof fairings • High-voltage components with warning labels			
2. Immobilization/Stabilization/Lifting			
		Height control in driver's compartment.	
		Seat height and forward/rear adjustment in driver's compartment.	
		Use only these lifting points.	 WARNING If the vehicle has to be lifted, never put lifting equipment under the rear batteries.
 		IMMOBILIZATION PROCEDURE: 1. Chock the wheels (if vehicle upright). 2. Set the parking brake (yellow knob, left of driver). 3. Place the vehicle in neutral (drive selector). 4. Turn the MASTER CONTROL SWITCH to the OFF position. 5. Use the door EMERGENCY AIR RELEASE valve to release air pressure from the door motors. Turn the door control lever (left of driver) to the F position to prevent doors from closing.	

	Identification number	Version number	Page number
	800077265	2024 August	2

3. Disable Direct Hazards/Safety Regulations

  	- The rated voltage on the LFSe+ vehicle is 655VDC. - It is possible to shut down the high-voltage system in two places.
	Tell-tales and a beeping sound indicate a thermal runaway detection in the lithium-ion batteries.
 MASTER CONTROL SWITCH	SWITCHING OFF THE IGNITION Turn the MASTER CONTROL SWITCH to the OFF position.

SHUT DOWN HIGH VOLTAGE:

Option 1 (Non-Destructive Method)

- Push the emergency stop button, located in the driver's compartment.
- Turn the **12/24V BATTERY DISCONNECT SWITCH** to the **OFF** position (if the streetside motor access door is accessible).

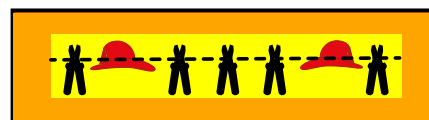


Option 2 (Destructive Method)

If the driver's area and streetside motor access door are inaccessible:

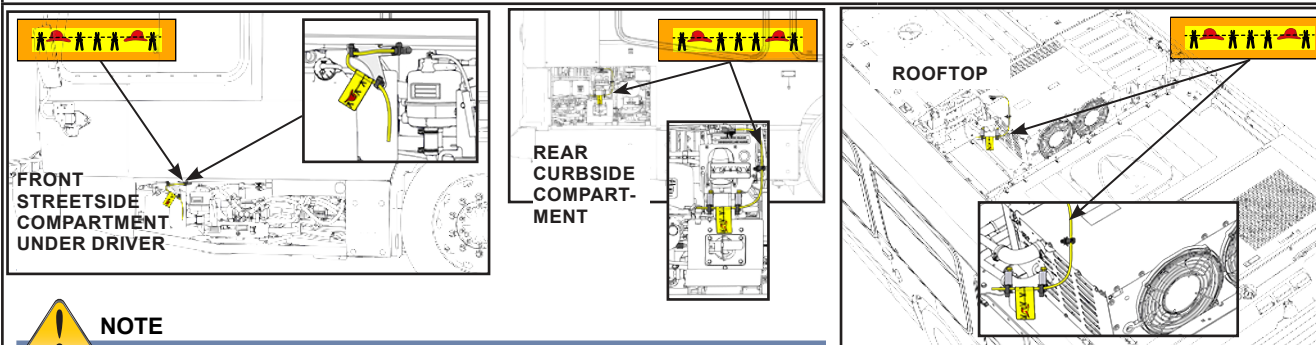
Cut the **YELLOW WIRE** at **ONE OF THE FOUR** locations:

- Inside streetside electronic steering compartment
- Inside curbside motor access door (curbside and streetside)
- Both sides of the BTMS, on the roof



WARNING

The high-voltage system contains enough energy to maintain an electric arc. Full respect of the safety procedures and proper use of the specified PPE rated for 25 cal/cm² reduces the risk of exposure to an electric arc.



NOTE

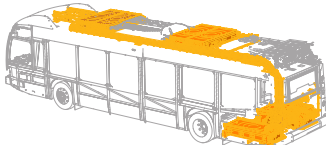
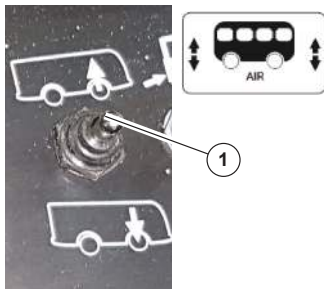
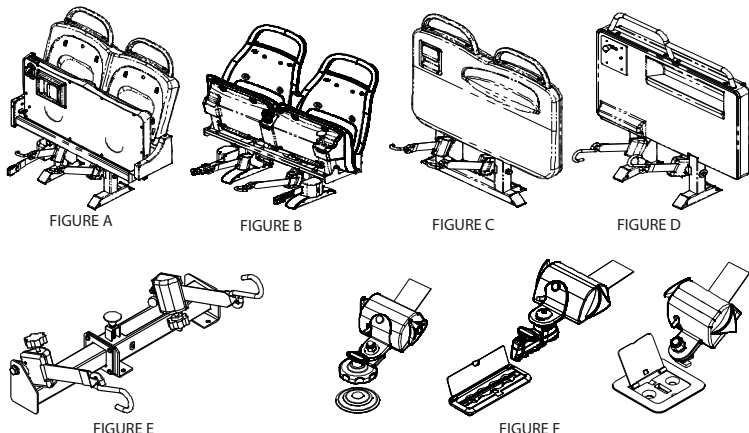
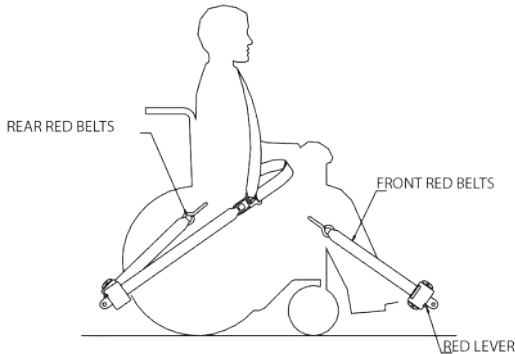
Cutting at any of the locations shown will disable the high-voltage supply.

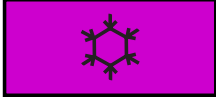
IF THE BUS IS CHARGING

- Open either the curbside or streetside motor access door.
- Press the black stop button on the CSS backplate. The three lights on the backplate will illuminate.
- Wait for the three lights to extinguish, then disconnect the charging plug.



	Identification number	Version number	Page number
	00-00-44	2024 August	3

4. Access to the Occupants			
		1 emergency exit roof hatch at rear 4 emergency exits: 2 push-out windows on each side	
		Two door exits	
		The materials used for the vehicle shell and body are composed of stainless steel (blue) and fiberglass (yellow).	
		The no-cut zones include any portion of the orange high-voltage (HV) cables, any of the four to six ESS HV battery packs, and any components contained in the rear motor compartment (including the low-power charging ports on either side of the vehicle). The prohibited cut zones are to be respected when the high voltage is active.	
			TO ADJUST ANGLE: 1. Pull the knob up. 2. Place the steering wheel in the desired position. 3. Release the knob. TO ADJUST HEIGHT: 1. Push the knob down. 2. Place the steering wheel in 
		To adjust the height: 1. Before discharging passengers via the front door, activate (down position) the front kneeling switch (item 1) to release air from the front suspension. The front of the vehicle will lower between 4 and 5 inches (10 and 13 cm). While the front is being lowered, an alarm sounds and a light flashes on the outside of the front door. Also, a tell-tale will light up to indicate that the operation is in progress. 2. To raise the vehicle, set the kneeling switch in the up position. Once the vehicle reaches the minimum operating height, control of the brakes and accelerator is restored.	
ACCESSIBILITY SEATING SECUREMENT			
FLOOR MOUNTED SECUREMENTS			
			
			
		Identification number	Version number
		00-00-149	2024 August
		Page number	
		4	

5. Stored Energy/Liquid/Gases/Solid						
						
						
						

HIGH-VOLTAGE (HV) BATTERY INFORMATION - GENERAL FIRST AID MEASURES AND ENVIRONMENTAL ASPECTS

Damage to one or more battery cells may cause uncontrolled increases in temperature and pressure (thermal runaway), which can lead to several possible hazards, as described below:

EXPOSURE TO HV: 60V DIRECT CURRENT (VDC) OR 30V ALTERNATING CURRENT (VAC)

- Avoid contact with HV cabling and components. ALWAYS assume the HV system is energized.
- Avoid contact with a damaged HV battery, a significant shock hazard may exist.
- NEVER cut orange HV cabling or penetrate or breach HV components with tools.
- The HV battery will remain energized after following HV system shutdown procedures in Section 3.
- Even with the HV batteries completely discharged (State of charge = 0), the system remains within the class B voltage definition and should, therefore, be disconnected as indicated in Section 2.
- Sparks, smoke, bulging, or bubbling noises coming from the HV battery are signs of a potentially overheating battery, which could result in a delayed fire.
- Follow local medical protocols and provide the First Aid required for any burn or other injuries.

EXPOSURE TO ELECTROLYTE MIXTURE:

- Wear appropriate PPE if exposure to electrolyte is expected. SCBA is highly recommended due to possible fumes.
- Clothing or PPE that may have contacted electrolyte should be decontaminated or discarded appropriately.
- Always contact medical assistance.

INHALATION IN NON-FIRE SITUATIONS:

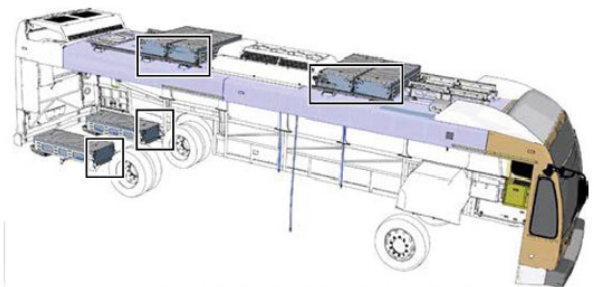
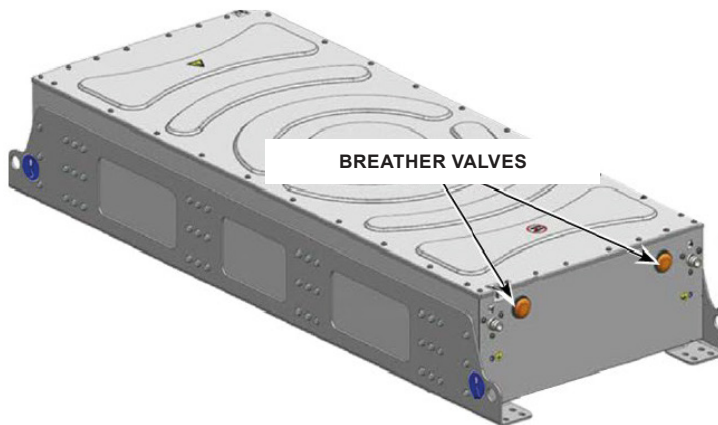
- If you detect leaking fluids, sparks, smoke, or bubbling noises coming from the HV battery, open windows to ventilate the vehicle and prevent the buildup of fumes. Electrolytic vapors may be toxic or severely irritating. If vapors are inhaled, immediately move to fresh air.

SPILL/LEAK HAZARDS FROM HV BATTERIES:

- HV Lithium-Ion batteries are considered dry cell batteries, and if damaged or breached, electrolyte leakage should be minimal.
- The HV batteries are liquid cooled with coolant flowing to and from an external radiator unit on the roof. If the cooling circuit is damaged or disconnected from the batteries or if the battery housing is breached, coolant may leak. The coolant is a water/glycol/ethylene solution, similar to that of the conventional vehicle radiators and should not be confused with battery electrolyte. Battery electrolyte is internal to the individual battery cells inside the battery pack.
- If damage is extensive, cross contamination with battery electrolyte is possible.

	Identification number	Version number	Page number
	00-00-149	2024 August	5

6. In case of fire		
		- Establish a 50-foot (15-meter) radius SAFETY ZONE around the vehicle. - Lithium-ion batteries contain fuel and oxidizer. - Lithium-ion batteries can self-ignite, and/or reignite even after a fire has been extinguished. Damaged batteries can vent toxic gas. Vehicles may be equipped with an optional automatic fire suppression system. DO NOT EXTINGUISH BATTERY FIRES WITH SMALL AMOUNTS OF WATER. Establish an additional water source as battery fires can take up to 24 hours to burn out and several thousand gallons of water. - Do not use ABC fire extinguishers to put out a battery fire. - Standard firefighting practices can be applied to other portions of the vehicle to prevent the fire from spreading.
		- If coolant escapes from the battery cooling system, there is a thermal reaction risk in the high-voltage batteries. Monitor the temperature (with a thermal imaging camera) of the high-voltage battery. Wear appropriate PPE. - During thermal runaway, the lithium-ion batteries can release hydrogen fluoride, which can be harmful. - Breather valves on the HV batteries may emit large flames as a result of thermal runaway.



BREATHING VALVE ORIENTATION

ROOFTOP BATTERIES: CURBSIDE
MOTOR COMPARTMENT: VALVES FACE TOWARD THE FRONT OF THE VEHICLE

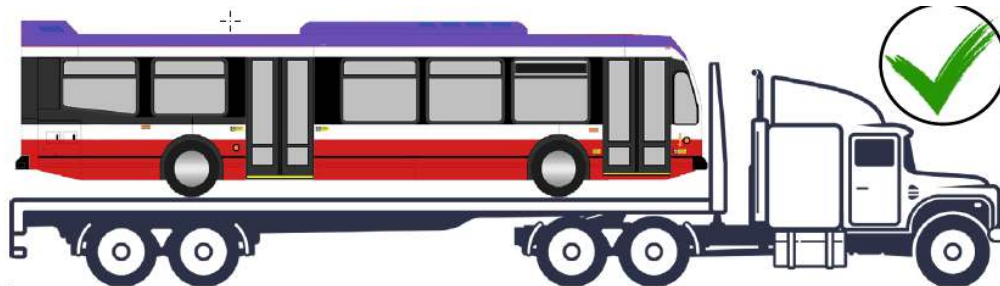
7. In case of water submersion	
	- The HV system is isolated from the chassis and is designed to pose no shock hazard from touching the vehicle body. - The system is designed not to energize surrounding water and is equipped with short-circuit fault detectors designed to shut down the HV system. - Avoid contact with HV components, cabling, or service disconnects. - Follow standard practices and procedures for passenger access and vehicle removal from water. - Wear appropriate PPE. - Follow heading 3. Disable direct hazards/safety regulations once out of water. Allow the water to drain.

	Identification number	Version number	Page number
	00-00-149	2024 August	6

8. Towing/Transportation/Storage



- If the batteries are defective, park the vehicle at least 50 feet (15 meters) from buildings and other vehicles (quarantine area). Risk of self-ignition or reignition. If the batteries are in good condition, park the vehicle at least 40 feet (12 meters) from buildings and other vehicles (quarantine area).
- Use thermal infrared camera to make sure the battery is not hot.
- **Do NOT** tow a vehicle that has been involved in an incident on its drive wheels. The electric motors can produce electricity when moving the vehicle with the rear drive wheels on the ground. This is a potential source of electric shock.
- The vehicle can be transported on a flatbed, preventing any wheel rotation.
- Observe the vehicle for a minimum of 48 hours using a thermal infrared camera or other method. Respond quickly and properly in case of thermal runaway.



MAXIMUM LOADING DURING LIFTING AND TOWING:

This information specifies the loading that can be applied when using a towing hook, towing hitch cross-member, axles and torque stay anchorages.

Single towing hook: Do not load the hook more than the vehicle gross weight.

Double towing hooks: Do not load each hook more than half the vehicle gross weight.

Towing hitch, towing Hitch Cross-Member: Max. 25.5 inches (648 mm) from center of member

MINIMUM JACK REQUIREMENTS

Weight Capacity 10 tonnes (10,000 kg)

Diameter of jack head..... 2 in. (51 mm)

MINIMUM SUPPORT REQUIREMENTS

Weight Capacity 10 tonnes (10,000 kg)

Diameter of support head 6 in. (152 mm)

TRACK

Aluminum Wheels

Front 86.8 in. (2,205 mm)

Rear 77.2 in. (1,960 mm)

Steel Wheels

Front 86.3 in. (2,191 mm)

Rear 76.5 in. (1,942 mm)

	Identification number	Version number	Page number
	00-00-149	2024 August	7